

Explicit Richardson Scheme

As a second explicit approach, the Richardson scheme can be applied to the one-dimensional transient diffusion equation. Unlike the FTCS scheme, which uses a forward difference in time, the Richardson scheme employs a central difference in time and a central difference in space.

The central temporal discretization provides second-order accuracy in time. However, despite its second-order accuracy in both time and space, the Richardson scheme is unconditionally unstable for the one-dimensional diffusion equation. Therefore, it has no practical value for the present transient Plane Couette Flow problem.

The governing diffusion equation is evaluated at the grid point (y_j, t^n) , giving:

$$\left(\frac{\partial u}{\partial t}\right)_j^n = \nu \left(\frac{\partial^2 u}{\partial y^2}\right)_j^n \quad (1)$$

The Richardson scheme uses:

- a second-order central difference in time;
- a second-order central difference in space;
- three consecutive time levels: $n - 1$, n , and $n + 1$.

A. Discretization of the First-Order Temporal Derivative

To approximate the first-order temporal derivative $\left(\frac{\partial u}{\partial t}\right)_j^n$, Taylor series expansions of u_j^{n+1} and u_j^{n-1} are developed about the point (y_j, t^n) .

Forward Temporal Taylor Expansion

The Taylor series expansion of $u(y_j, t^{n+1})$ about (y_j, t^n) is:

$$u_j^{n+1} = u_j^n + \left(\frac{\partial u}{\partial t}\right)_j^n \Delta t + \frac{1}{2!} \left(\frac{\partial^2 u}{\partial t^2}\right)_j^n (\Delta t)^2 + \frac{1}{3!} \left(\frac{\partial^3 u}{\partial t^3}\right)_j^n (\Delta t)^3 + \frac{1}{4!} \left(\frac{\partial^4 u}{\partial t^4}\right)_j^n (\Delta t)^4 + \dots \quad (2)$$

Backward Temporal Taylor Expansion

Similarly, the Taylor series expansion of $u(y_j, t^{n-1})$ about (y_j, t^n) is:

$$u_j^{n-1} = u_j^n - \left(\frac{\partial u}{\partial t}\right)_j^n \Delta t + \frac{1}{2!} \left(\frac{\partial^2 u}{\partial t^2}\right)_j^n (\Delta t)^2 - \frac{1}{3!} \left(\frac{\partial^3 u}{\partial t^3}\right)_j^n (\Delta t)^3 + \frac{1}{4!} \left(\frac{\partial^4 u}{\partial t^4}\right)_j^n (\Delta t)^4 - \dots \quad (3)$$

Subtracting Equation (3) from Equation (2) eliminates the even-order temporal derivative terms:

$$u_j^{n+1} - u_j^{n-1} = 2 \left(\frac{\partial u}{\partial t} \right)_j^n \Delta t + \frac{1}{3} \left(\frac{\partial^3 u}{\partial t^3} \right)_j^n (\Delta t)^3 + \mathcal{O}((\Delta t)^5) \quad (4)$$

Rearranging Equation (4) to isolate the first-order temporal derivative gives:

$$\left(\frac{\partial u}{\partial t} \right)_j^n = \frac{u_j^{n+1} - u_j^{n-1}}{2\Delta t} + \mathcal{O}((\Delta t)^2) \quad (5)$$

The central difference approximation is therefore second-order accurate in time. The temporal truncation error is $\mathcal{O}((\Delta t)^2)$.

B. Discretization of the Second-Order Spatial Derivative

For the second-order spatial derivative, the same central difference approximation used in the FTCS scheme is applied. The Taylor expansions about (y_j, t^n) lead to:

$$\left(\frac{\partial^2 u}{\partial y^2} \right)_j^n = \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{(\Delta y)^2} + \mathcal{O}((\Delta y)^2) \quad (6)$$

The spatial approximation is second-order accurate because its truncation error is $\mathcal{O}((\Delta y)^2)$.

C. Discrete Algebraic Equation

Substituting the central temporal approximation from Equation (5) and the central spatial approximation from Equation (6) into the governing equation in Equation (1) gives:

$$\frac{u_j^{n+1} - u_j^{n-1}}{2\Delta t} = \nu \left(\frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{(\Delta y)^2} \right) \quad (7)$$

Multiplying both sides of Equation (7) by $2\Delta t$ yields:

$$u_j^{n+1} - u_j^{n-1} = \frac{2\nu\Delta t}{(\Delta y)^2} (u_{j+1}^n - 2u_j^n + u_{j-1}^n) \quad (8)$$

Rearranging Equation (8) to isolate the unknown value u_j^{n+1} gives:

$$u_j^{n+1} = u_j^{n-1} + \frac{2\nu\Delta t}{(\Delta y)^2} (u_{j+1}^n - 2u_j^n + u_{j-1}^n) \quad (9)$$

The dimensionless diffusion number is defined as:

$$d = \frac{\nu \Delta t}{(\Delta y)^2} \quad (10)$$

Using the diffusion number defined in Equation (10), the Richardson scheme becomes:

$$u_j^{n+1} = u_j^{n-1} + 2d(u_{j+1}^n - 2u_j^n + u_{j-1}^n) \quad (11)$$

Equation (11) is the final explicit Richardson update equation. It calculates the unknown value at time level $n + 1$ using the known solution values at time levels n and $n - 1$.

For a grid containing N nodes, Equation (11) is applied at all interior nodes:

$$j = 1, 2, \dots, N - 2$$

The boundary values at $j = 0$ and $j = N - 1$ are assigned directly from the prescribed physical boundary conditions.

Because Equation (11) uses two previous time levels, the Richardson method is classified as an explicit three-time-level scheme. No algebraic system needs to be solved at each time step.

D. Accuracy of the Richardson Scheme

According to Equation (5), the temporal discretization is second-order accurate:

$$\text{Temporal truncation error} = \mathcal{O}((\Delta t)^2)$$

According to Equation (6), the spatial discretization is also second-order accurate:

$$\text{Spatial truncation error} = \mathcal{O}((\Delta y)^2)$$

Therefore, the Richardson scheme is second-order accurate in both time and space. Its combined local truncation error can be expressed as:

$$\mathcal{T}_{\text{Richardson}} = \mathcal{O}((\Delta t)^2) + \mathcal{O}((\Delta y)^2) \quad (12)$$

Although this formal accuracy is higher than the first-order temporal accuracy of the FTCS scheme, accuracy alone does not guarantee numerical stability.

E. Stability Analysis of the Explicit Richardson Scheme

Despite its second-order accuracy in both time and space, the Richardson scheme is unconditionally unstable for the one-dimensional diffusion equation.

This behavior can be demonstrated using a von Neumann Fourier stability analysis. Consider a Fourier error mode of the form:

$$\epsilon_j^n = G^n e^{ij\theta} \quad (13)$$

where G is the amplification factor, θ is the phase angle, and $i = \sqrt{-1}$.

Substituting the Fourier mode from Equation (13) into the Richardson update equation in Equation (11) gives:

$$G^{n+1} e^{ij\theta} = G^{n-1} e^{ij\theta} + 2dG^n (e^{i(j+1)\theta} - 2e^{ij\theta} + e^{i(j-1)\theta}) \quad (14)$$

Dividing Equation (14) by $G^{n-1} e^{ij\theta}$ gives:

$$G^2 = 1 + 2dG (e^{i\theta} - 2 + e^{-i\theta}) \quad (15)$$

Using the identities $e^{i\theta} + e^{-i\theta} = 2 \cos \theta$ and $\cos \theta - 1 = -2 \sin^2 \left(\frac{\theta}{2} \right)$, Equation (15) becomes:

$$G^2 + 8d \sin^2 \left(\frac{\theta}{2} \right) G - 1 = 0 \quad (16)$$

The two roots of the characteristic equation in Equation (16) are:

$$G_{\pm} = -4d \sin^2 \left(\frac{\theta}{2} \right) \pm \sqrt{1 + 16d^2 \sin^4 \left(\frac{\theta}{2} \right)} \quad (17)$$

For every $d > 0$ and every nonzero wave number, one of these roots has a magnitude greater than unity. In particular, the negative root satisfies:

$$|G_-| = 4d \sin^2 \left(\frac{\theta}{2} \right) + \sqrt{1 + 16d^2 \sin^4 \left(\frac{\theta}{2} \right)} > 1 \quad (18)$$

Therefore, at least one numerical error mode grows with successive time levels for any positive diffusion number d . The Richardson scheme is consequently unconditionally unstable:

$$|G_-| > 1 \quad \text{for } d > 0 \text{ and } \sin \left(\frac{\theta}{2} \right) \neq 0 \quad (19)$$

This result means that reducing the time step cannot eliminate the instability. Even when d is very small, one of the two amplification-factor roots remains greater than unity in magnitude for at least one nonzero Fourier mode.

Consequently, the Richardson scheme produces growing numerical disturbances and eventually leads to non-physical solutions. It is therefore unsuitable for practical transient diffusion simulations, including the present Plane Couette Flow problem.

F. Summary of the Richardson Scheme

The main properties of the explicit Richardson scheme are summarized as follows:

- It uses central differencing in time.
- It uses central differencing in space.
- It is second-order accurate in time.
- It is second-order accurate in space.
- It is an explicit three-time-level method.
- It is unconditionally unstable for the one-dimensional diffusion equation.
- It is not suitable for practical transient diffusion calculations.

Therefore, the Richardson scheme is included in this report as an important numerical example. It demonstrates that a scheme may possess a high formal order of accuracy while still being unsuitable because of its stability characteristics.